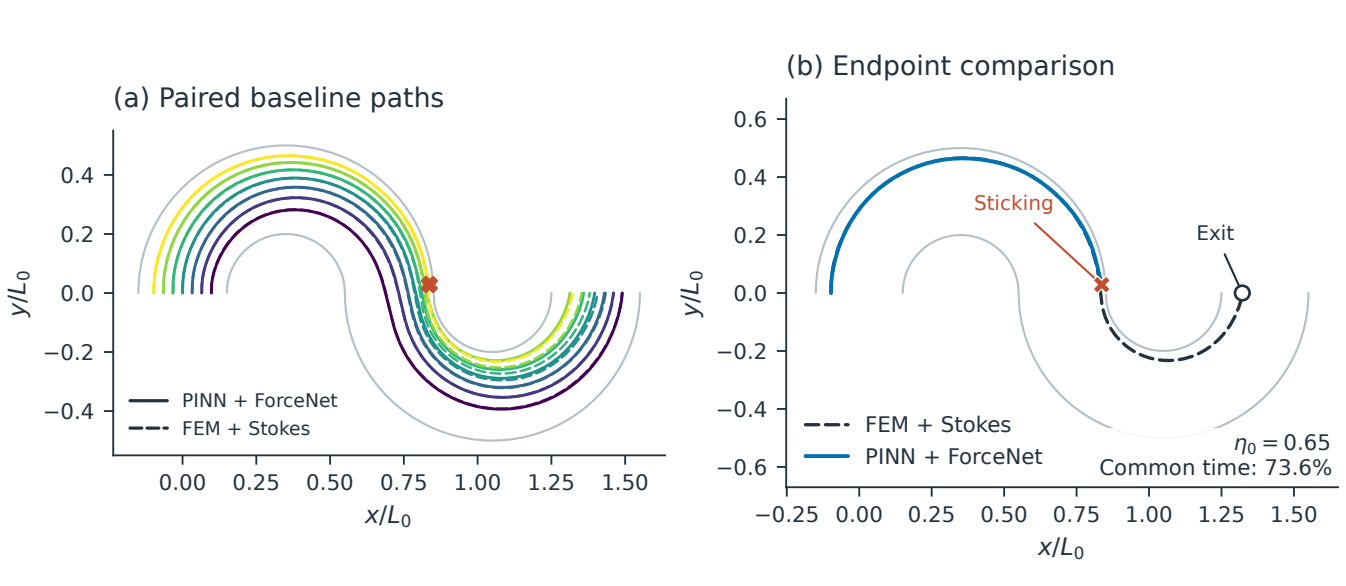
(a) Paired baseline paths
PINN + ForceNet
FEM + Stokes
y/L_0
x/L_0
0.4
0.2
0.0
−0.2
−0.4
0.00
0.25
0.50
0.75
1.00
1.25
1.50
(b) Endpoint comparison
Sticking
Exit
FEM + Stokes
PINN + ForceNet
$\eta_0 = 0.65$
Common time: 73.6%
y/L_0
x/L_0
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50